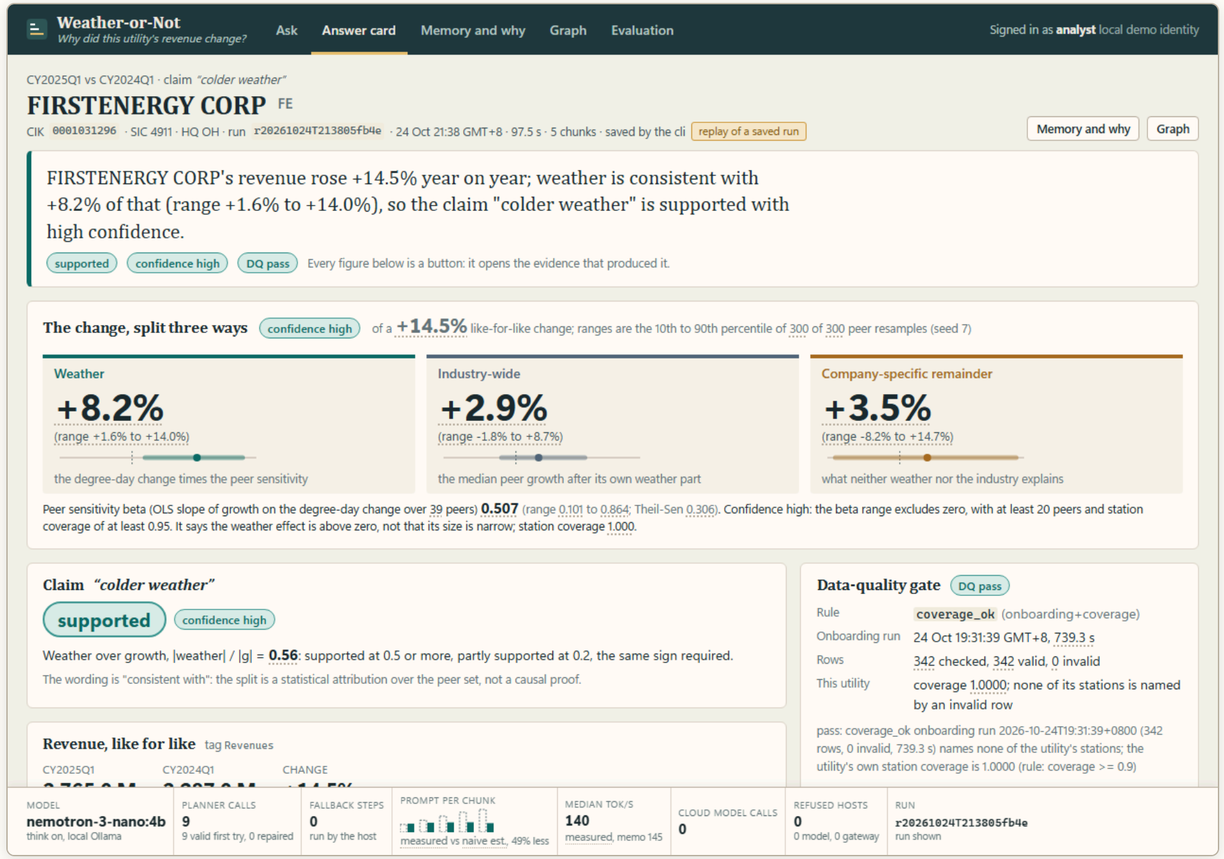
# Weather-or-Not

**The utility says it was the weather. Weather-or-Not checks.**
It splits a US electric utility's year-on-year revenue change into weather, industry-wide and company-specific parts, each a range, with the Lightning SQL behind every number.

| | |
|---|---|
| Zetaris | The only data path: SEC EDGAR, NOAA GHCN-Daily parquet and company_dns read through Lightning SQL; degree days computed in SQL; a USL with keys and a data-quality gate; the SQL text is the citation. |
| Meterless | H-MEM facts with Zetaris provenance and recast conflicts, a Markovian question chain with a flat carryover, World Model edges with validity dates, Scout-style signed contracts. |
| NVIDIA | Nemotron 3 Nano 4B through local Ollama picks each next step the checklist allows and writes the memo; it never computes a number. 0 cloud model calls. |

Umar Mohktar, Singapore - solo




The answer card for FirstEnergy, CY2025Q1 against CY2024Q1: saved run of 24 Oct 2026, rendered by the Weather-or-Not UI's card view. Every figure is a button that opens the evidence behind it.

**FirstEnergy, Q1 2025: the company says heating degree days +15%; the agent's 9 stations say +16.2%; weather is consistent with 8.2 of the 14.5 points of revenue growth.**

Data: SEC EDGAR (US government work, public domain; queried live, never redistributed), NOAA GHCN-Daily (CC0-1.0), company_dns SIC list (Apache-2.0 service).

# "It was the weather" is easy to say and slow to check

## The users: anyone handed that sentence who has to decide whether to believe it

A utility credit analyst in earnings season, a state commission staff analyst or consumer advocate preparing for a rate case, a reporter. Each gets a filing that credits the weather and has hours, not days, to test it.

## The problem: the claim is public, the check is not

FirstEnergy's Q1 2025 10-Q (accession 0001031296-25-000017) says heating degree days were **"15% above the same period of 2024"**. Testing that by hand means degree days for dozens of stations, revenue for every peer, and the comparative column of each filing, which is **recast** between filings, so a naive year-on-year reads the wrong number.

**43**

seed utilities whose states and airport stations are curated from public filings

56 GHCN stations in 43 states

**1,937**

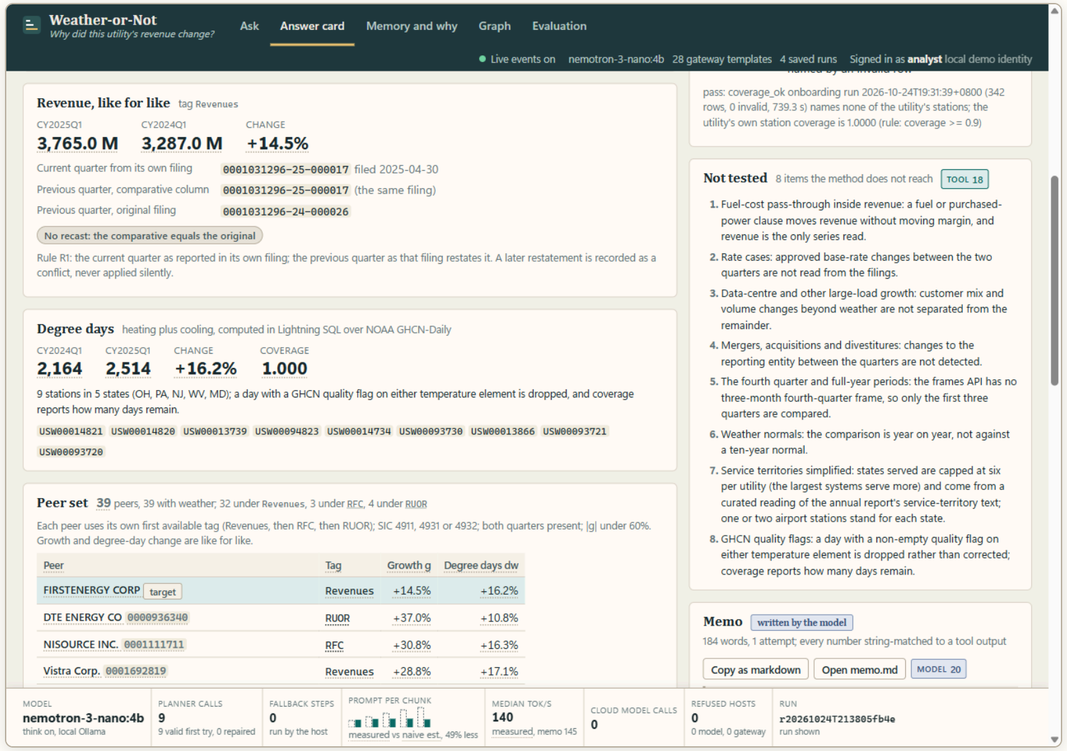
filers in one SEC frames period (Revenues, CY2025Q1); the peer set is cut from three such tags

count equal to the source response

**+15% / +16.2%**

heating degree days in FirstEnergy's own 10-Q against total degree days at the agent's 9 stations: the one outside check

10-Q accession 0001031296-25-000017




What the analyst sees instead of a sentence: both accession numbers, the like-for-like rule, degree days over the utility's nine stations, and the 39-peer table, each number cited.

Weather-or-Not answers in about two minutes with ranges, a claim verdict, a "Not tested" list, and the SQL behind every number.

Quote from the FirstEnergy Q1 2025 10-Q, accession 0001031296-25-000017 (SEC EDGAR). Counts from seeds/utilities.json, seeds/stations.json and zetaris/README.md; the agent's +16.2% from the saved runs (docs/eval-report.md section 4).

# Why a naive year-on-year can read the wrong number, and rule R1

The SEC frames API serves the **most recently filed** fact for a period. A quarter's value can be restated in a later 10-Q's comparative column, so the number a script reads today is not the number the company reported in its own quarter.

**AEP's own Q1 2025 10-Q**

**5,463.4 M**

`0000004904-25-000061, filed 2025-05-06`

in use, rule R1

against

**The Q1 2026 10-Q, comparative column**

**5,463.0 M**

`0000004904-26-000034, filed 2026-05-05`

recorded as conflict rc-4cd5f8bdf0bd

### Rule R1, fixed in advance

Use the current quarter's value as reported in its **own** filing, and the previous quarter as that same filing restates it (like-for-like). A later restatement is recorded in H-MEM as a `recast_comparative` conflict with both values and both accession numbers, never applied silently. The analyst can override it with a signed high-risk contract; the override and its reason go into the trust ledger.

On AEP the restatement is 0.4 M on 5,463 M and moves nothing on the card; the rule matters when a restatement is material, as on the synthetic fixture (+12.2% like for like against +10.0% naive).

**128 of 133**

rows of the USL `revenue_quarter` table are comparatives, not first filings

what the frames endpoint serves today

**+12.2% vs +10.0%**

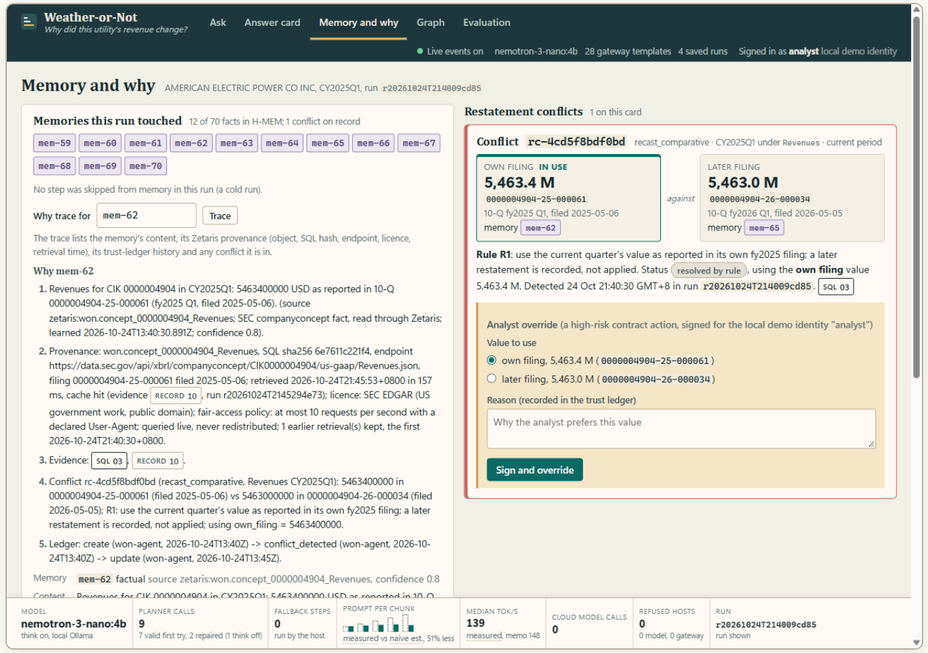
like-for-like against naive growth on the synthetic recast fixture

agent/test/fixtures, labelled synthetic

**3,287.0 M**

FirstEnergy's CY2024Q1 comparative equals its original: no recast, and the card says so

0001031296-24-000026 and -25-000017

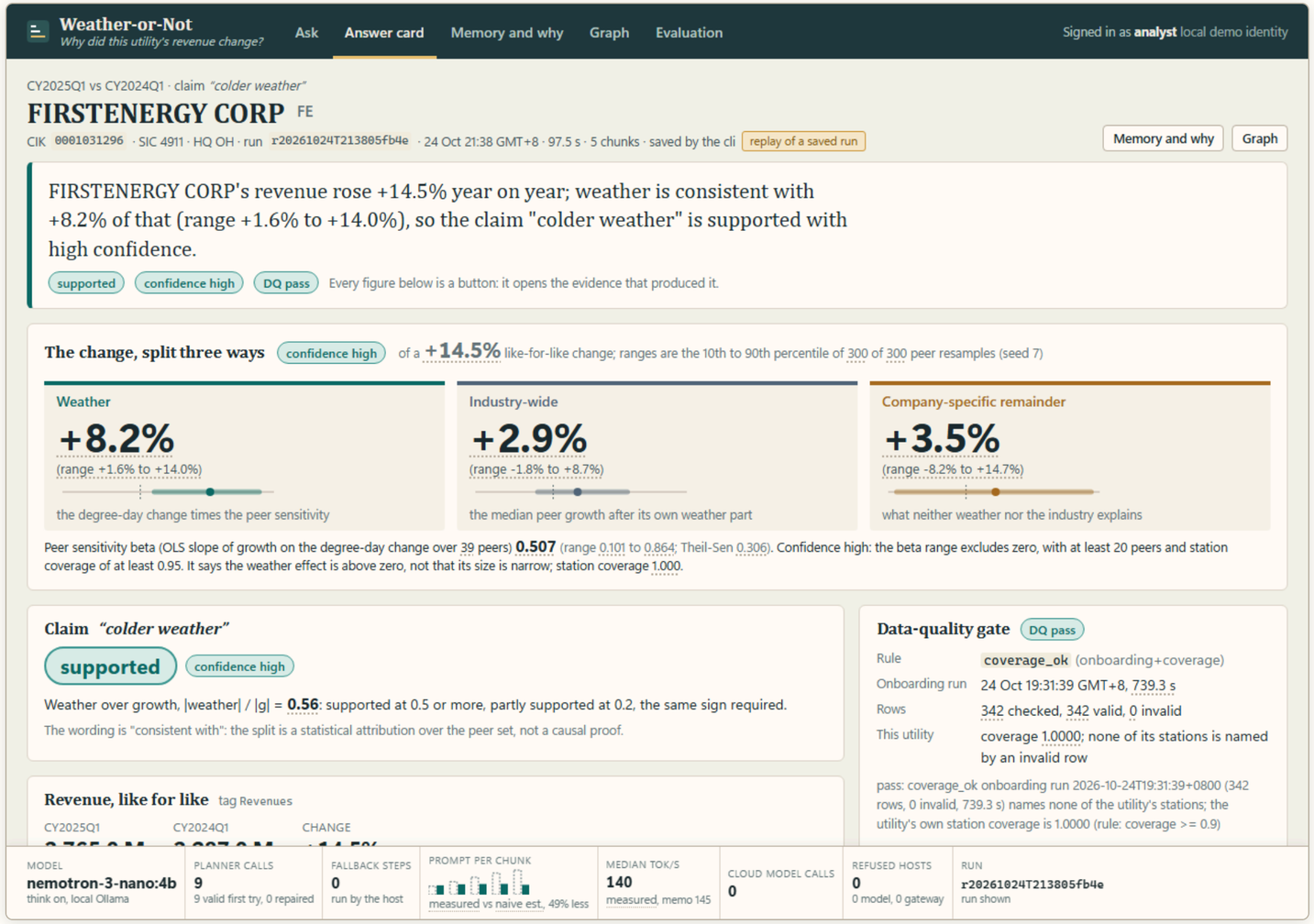



The live conflict on the Memory and why screen: own filing against later filing, rule R1, the why trace of memory mem-62 (object, SQL hash, endpoint, licence, retrieval time, ledger) and the signed analyst override.

Values read through Zetaris from SEC companyconcept (won.concept_0000004904_Revenues), run r20261024T214009cd85; USL counts from zetaris/README.md; the synthetic fixture from agent/test/fixtures.

THE ANSWER

# FirstEnergy, CY2025Q1: weather is consistent with 8.2 of the 14.5 points, somewhere between 1.6 and 14.0




Run r20261024T213805fb4e (cold, 5 chunks, 97.5 s). The verdict sentence, the split, the claim test and the DQ gate; the telemetry footer below belongs to the same run.

| Weather | Industry-wide | Remainder |
|---|---|---|
| **+8.2%** | **+2.9%** | **+3.5%** |
| range +1.6% to +14.0% | range -1.8% to +8.7% | range -8.2% to +14.7% |

of a **+14.5%** like-for-like change (3,765.0 M against 3,287.0 M); weather and industry ranges are the 10th to 90th percentile of 300 of 300 valid peer resamples (seed 7); the remainder range combines their bounds, so it is wider.

| | |
|---|---|
| Claim "colder weather" | **supported** ratio 0.56 |
| Confidence | high |
| Degree days, 9 stations in 5 states | 2,164 to 2,514, +16.2% |
| Peer sensitivity beta over 39 peers | 0.51 (range 0.10 to 0.86) |
| Theil-Sen cross-check | 0.31 |
| Peers by tag; excluded | 32 + 3 + 4; 3 |
| Station coverage; DQ gate | 1.00; pass |
| Memo by the model, numbers string-matched | 184 words, 1 attempt, 0 unknown |
| Not tested, listed on the card | 8 items |

**Supported** because the central estimate is over half the change (ratio 0.56). **High**: the beta range excludes zero with 39 peers and coverage 1.00, so the weather effect is above zero; it does not say 8.2 is precise.

The wording is "consistent with": a statistical attribution over the peer set, not a causal proof.

Every number from data/runs/r20261024T213805fb4e/run.json as rendered by the UI and tabulated in docs/eval-report.md section 4; identical across the five FirstEnergy runs.

AGENT WORKFLOW

# The model plans and writes; the host runs the tools and computes

**Ask**
Utility (CIK), quarter, management's claim. The question becomes a Markovian chain: at most 10 chunks, 2 tool calls per chunk, 800 tokens of carried state.

›

**Plan**
Nemotron returns JSON against a schema: a thought, one tool from the allowed set, and its state. One repair round, then the host takes the next checklist step itself.

›

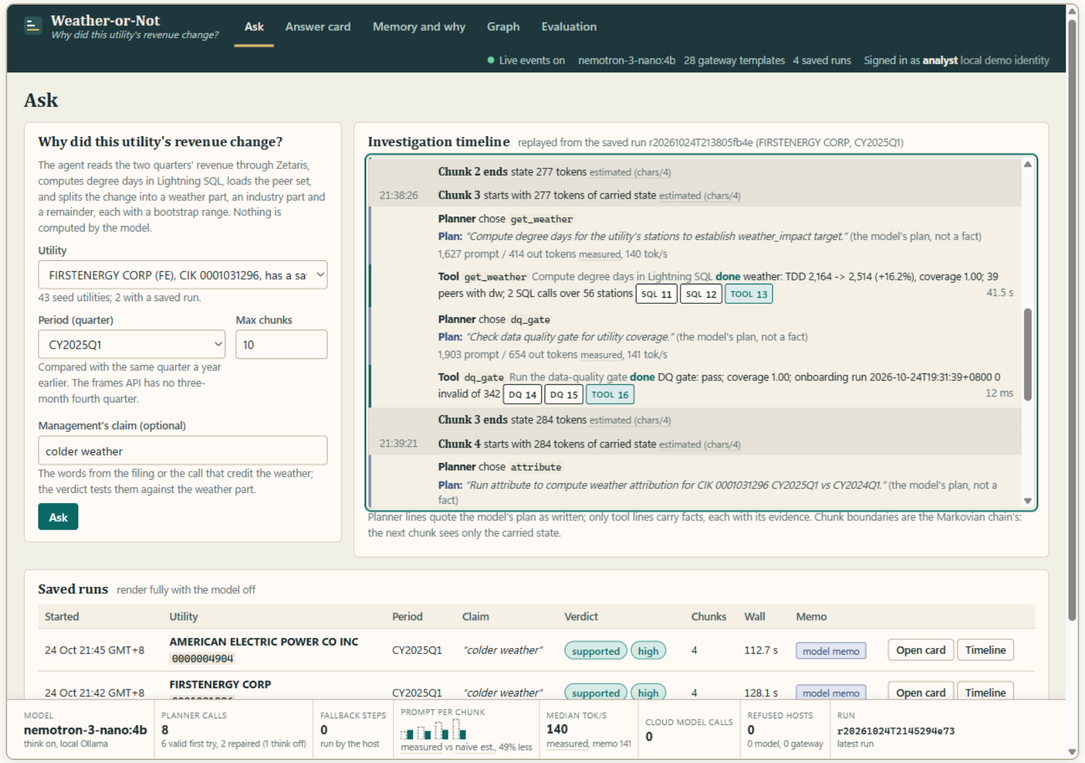
**Run the tool**
Ten host-run tools; each records evidence (SQL text, sha256, rows, object, endpoint, licence, latency, cache state) and returns a digest of at most 800 tokens.

›

**Check**
An 11-item checklist in fallback order; the DQ gate can withhold the verdict; every card claim must cite a recorded evidence id.

›

**Write**
The memo's every number is string-matched against the tool outputs; a draft with any other number is rejected once, then a template memo is used.




Run r20261024T213805fb4e replayed in the UI: planner lines are quoted as the model's plan, never as a fact; tool lines carry the facts and their evidence chips; chunk boundaries show the carried state.

### The ten tools

`resolve_utility`, `probe_revenue_tag`, `register_frames`, `get_revenue`, `get_peers`, `get_weather`, `dq_gate`, `attribute`, `not_tested`, `finalize`. Degree days run in Lightning SQL; growth, peers, rule R1 and the bootstrap attribution run in deterministic TypeScript.

### The checklist

utility_resolved, tag_chosen, revenue_like_for_like, peers_loaded, weather_target, weather_peers, dq_checked, attribution_done, claim_tested, not_tested_listed, memo_written. A warm run skips `tag_chosen` from memory: 4 chunks instead of 5. The planner's pick was the checklist's next step in 33 of 34 calls on the four clean runs and 34 of 36 on four other utilities; the exceptions are allowed reorderings that leave the card unchanged.

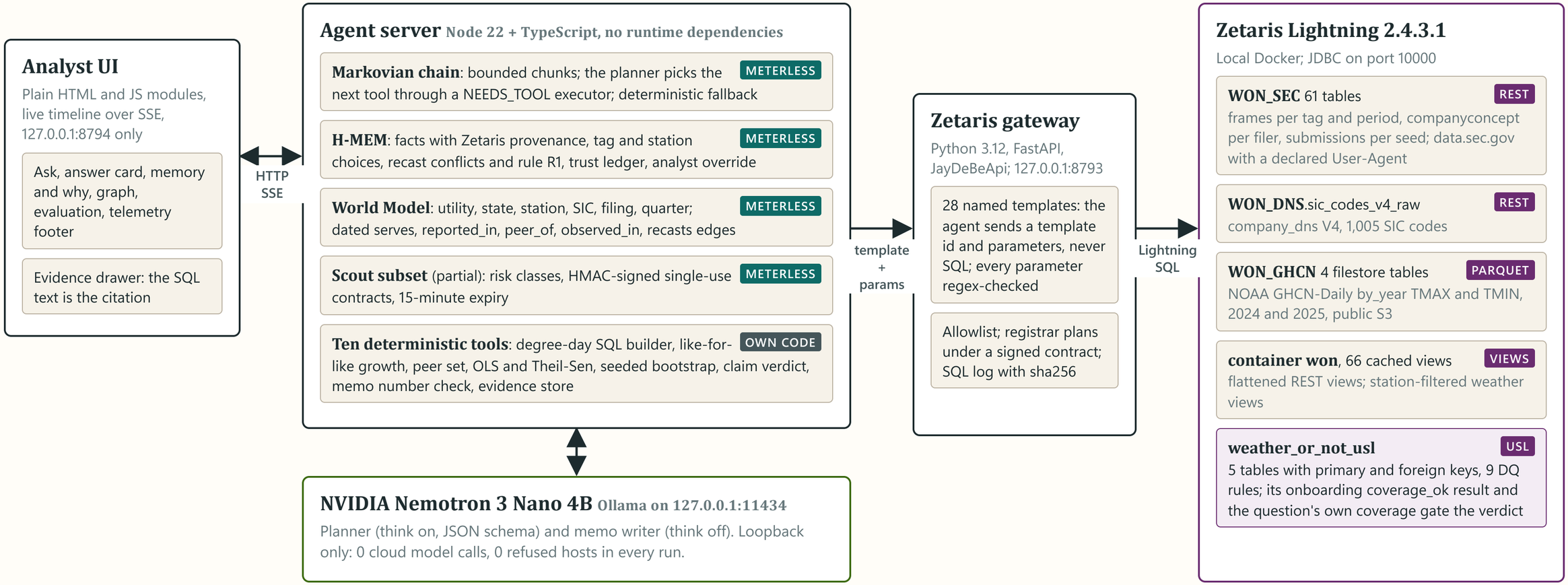
**9 / 9**
planner replies valid first try, FirstEnergy cold run

**0 of 12**
memo drafts with a number that is not a tool output

Tool and checklist names from agent/src/llm/schemas.ts; chain limits and the memo rule from docs/nemotron-bench.md; planner counts from data/runs/r20261024T213805fb4e/run.json.

ARCHITECTURE

# One machine, one data path: every source read goes through Zetaris




At runtime the agent never calls SEC EDGAR, NOAA or company_dns directly; only Zetaris does. Ports: UI 8794, gateway 8793, Ollama 11434, Zetaris JDBC 10000. Object counts from zetaris/README.md.

TECH STACK

# What it is built from, with the versions that ran

| Layer | What |
|---|---|
| Data platform | **Zetaris Lightning 2.4.3.1**, local Docker, JDBC `localhost:10000`; REST tables, Parquet filestore tables, SchemaStore views with CACHE, a USL with PK, FK and a custom DQ rule, LIST / RUN / SHOW DQ |
| Gateway | **Python 3.12.0**, FastAPI, JayDeBeApi over the Zetaris JDBC driver; 28 named templates, allowlist, registrar plans, execution-contract check; pytest 236 passed, 2 skipped |
| Agent | **Node 22.13.1 + TypeScript 7.0.2** run with tsx 4.23.15, no runtime dependencies; vitest 5.0.3, 16 files, 203 tests; `tsc --noEmit` 0 errors |
| Reasoning engines | **Meterless** H-MEM, Markovian and World Model reference engines, vendored unchanged at upstream commit `1742dec5` (Apache-2.0); Scout Intent subset reimplemented in `agent/src/contract` and labelled partial |
| Model | **NVIDIA Nemotron 3 Nano 4B** (`nemotron-3-nano:4b`) through Ollama on 127.0.0.1:11434; temperature 0, seed 7, num_ctx 8192, JSON schema output; think on for the planner (output budget 3,072), off for the memo (1,024) |
| UI | Plain HTML, CSS and ES modules, no build step, served by the agent server; server-sent events for the live timeline; light and dark; keyboard-reachable evidence drawer |
| Sources | SEC EDGAR frames, companyconcept and submissions; NOAA GHCN-Daily parquet (by_year, TMAX and TMIN); company_dns V4 SIC list. Seeds: 43 utilities, 56 stations, curated from public filings and labelled as such |
| Development | Developed in Cursor. |

### Why a 4B model is enough here

The model chooses a tool from a dynamic enum and writes prose. Every number comes from SQL or deterministic TypeScript, so the model's job is small enough for a local 4B model at about 140 tokens per second.

### Why no runtime dependencies

The agent is Node 22 standard library plus the vendored engines; the gateway is the only place with third-party packages. A fresh clone runs `npm ci`, `npm test` and the UI's empty state without Zetaris.

### Shared instance etiquette

Only `WON_*` databases, the `won` container and the `weatherornot` namespace are touched; never DROP, GRANT or INVALIDATE CACHE from code; SEC calls at most 10 a second with a declared User-Agent.

Versions from agent/package.json, engines/UPSTREAM.md, docs/nemotron-bench.md and the fresh-clone check of 24 Oct 2026 (Node, npm, Python, test counts; agent tests re-run 25 Oct 2026).

# Zetaris: data harness, degree-day engine, data-quality gate and citation

### 1 Objects per source, counted against the source

**WON_SEC** (REST): 8 frames tables, 10 companyconcept tables, 43 submissions profiles; every count equals the source response (frames Revenues CY2025Q1: 1,937 = 1,937). **WON_DNS**: 1,005 SIC codes. **WON_GHCN** (Parquet on public S3): 4 partitions of 4.31 to 4.37 million rows each, station-filtered views cached at 264 to 269 KB.

### 2 Degree days computed in Lightning SQL

One `weather_quarter` statement per quarter joins TMAX to TMIN over all 56 stations, drops days with a quality flag, and returns HDD, CDD, days observed and coverage in 18 to 24 s. Parity checked three ways on one station-quarter: 2022.69 HDD / 22.08 CDD from the tool's SQL, the template and the TypeScript reference.

### 3 The USL is the governed copy; its coverage_ok result is the gate

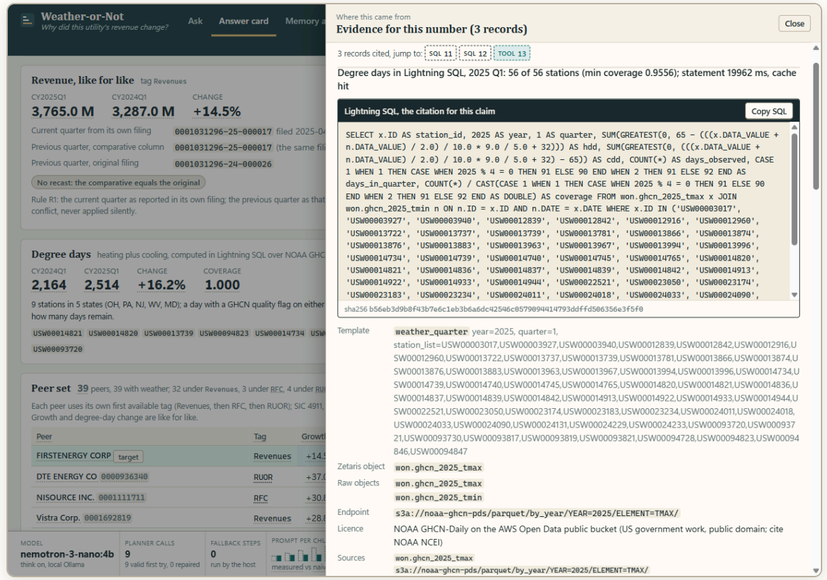
`weather_or_not_usl`: 5 tables (sic_code 1,005 rows, utility 43, station 57, weather_quarter 342, revenue_quarter 133), 5 primary keys, 3 foreign keys, custom rule `coverage_ok`. RUN DQ: 342 of 342 station-quarters valid; the utility FK rule found **1 invalid row**, MGE Energy under SIC 4900, which company_dns does not list: a real orphan found at onboarding; the peer filter excludes MGE by SIC code.

### 4 The SQL text is the citation; the agent never writes SQL

28 named templates behind an allowlist; a new frames period is registered at run time through the gateway's registrar under a signed contract (measured on an operator call: table, view, DESCRIBE, CACHE in 1.08 s; count 2,018 = source). Every card number opens the statement that produced it, with its sha256, object, endpoint, licence, retrieval time and cache state.

### 5 Join map, measured in Lightning SQL

Frames `cik` joins the submissions profile only through LPAD (34, 35, 21, 23, 9 and 11 matches across the six tag-periods, equal to the integer IN counts; 0 without LPAD); frames `loc` equals the business-address state in every row; 56 of 56 stations carry the seed file's state.




Evidence record 11 of the FirstEnergy run: the degree-day Lightning SQL for 2025 Q1 over 56 stations (19,962 ms, cache hit), its sha256, the template parameters, the Zetaris objects, the endpoint and the licence.

Counts and timings from zetaris/README.md and docs/eval-report.md (sections 1 and 5). Reads use the admin login behind the gateway's read-only allowlist; no least-privilege reader yet.

# Meterless engines hold the memory, bound the reasoning, date the world

### H-MEM: provenance, conflicts and the why trace

Vendored reference, subclassed. Every fact carries Zetaris provenance (object, SQL sha256, endpoint, accession, filing date, licence, retrieval time, latency, cache state, evidence id); a reconfirmed fact refreshes it and keeps the history: 70 facts after two questions, 16 provenance entries, 0 placeholders. The AEP restatement is conflict `rc-4cd5f8bdf0bd` (mem-62 against mem-65, rule R1); `why mem-62` prints the fact, provenance, evidence, conflict and ledger (create, conflict_detected, update).

### Markovian: a flat carryover instead of a growing history

Vendored reference generator. The carried state stayed at or below 284 tokens (estimated) against the 800 budget; measured first-call prompts stayed between 1,421 and 1,644 tokens per chunk while a full-history prompt (estimated, chars/4) would have grown to 3,742: 6,615 against 12,956 over the run, 48.9% less.

### World Model: what we knew, and since when

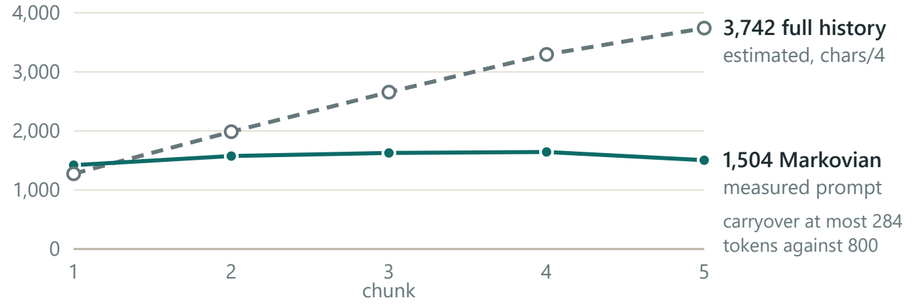
Vendored reference. 239 nodes (91 filings, 56 stations, 43 states, 43 utilities, 4 SIC codes, 2 quarters), 526 edges: `serves` valid from 2024-01-01 on the FY2024 10-K text, `reported_in` from the filing date, `peer_of` per period with g and dw, `observed_in` with HDD, CDD and coverage, one `recasts` edge. 769 events before and after the warm run: re-ingesting appends nothing.

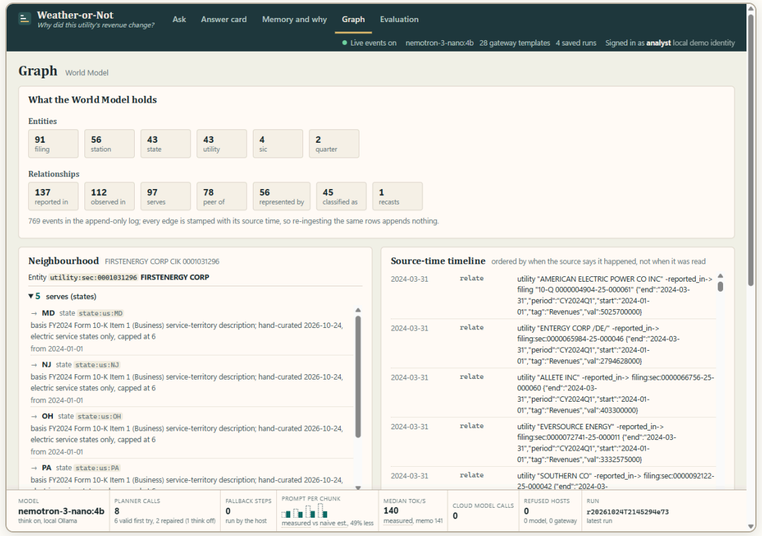
### Scout subset: signed, single-use contracts (partial)

Reimplemented from the spec. Registrar actions are medium risk; the analyst's `override_recast` is high risk. Contracts are HMAC-signed, expire after 15 minutes, and a replayed one is refused; the applied override recomputed growth with 5,463.0 M and the ledger records who, when and why.

### Prompt tokens per chunk, FirstEnergy cold run

Measured: Ollama prompt_eval_count of each chunk's first planner call. Estimated: full history, chars/4.




The Graph screen: World Model counts, FirstEnergy's neighbourhood with each edge's validity date, and the source-time timeline (406 entries).

WHERE NVIDIA CONTRIBUTES, MEASURED

# Nemotron 3 Nano 4B runs every model call locally, and it is measured

**139 to 140 tok/s**
median planner speed per run; memo 140 to 149 tok/s
Ollama eval counters, four clean runs

**28 / 34**
planner replies valid JSON on the first try; 6 repaired, 4 of them with think off; 0 fallbacks
four clean runs, 24 Oct 2026

**4 / 4**
memos kept on the first draft (184, 176, 184, 176 words), 0 numbers that were not a tool output
string-matched by the host

**0**
cloud model calls; 0 refused hosts: the client accepts loopback URLs only
counted in every run

### Planner ablation (25 Oct): FirstEnergy and AEP, cold then warm, per mode, each on an empty memory store

| Planner mode | Wall, mean | Valid first try | Repaired | Fallback | Output tokens | Model memos | Peak GPU memory |
|---|---|---|---|---|---|---|---|
| Think on (shipped) | 103.2 s | 27 / 34 | 6 | 1 | 33,225 | 4 / 4 | 7,145 MiB |
| Think off | 72.5 s | 27 / 34 | 7 | 0 | 14,595 | 4 / 4 | 6,981 MiB |
| No model | 38.5 s | n/a | n/a | every step | 0 | template | 3,765 MiB |

Wall ranges 94.0 to 111.2 s, 69.0 to 75.3 s and 38.0 to 39.4 s. With no model the host runs every checklist step and writes the template memo. All 12 cards are identical to the published ones: the numbers never come from the model. Think off was faster and no less reliable here; the shipped setting stays think on (frozen for judging). GPU memory is the whole card (nvidia-smi), 3,745 MiB with no model loaded. The video's 96 tok/s was recorded under screen capture; 139 to 140 tok/s is from the clean runs without capture.

### Stated plainly

Four of the six invalid first replies were empty: the model spent the whole output budget thinking on the `not_tested` and `finalize` steps. The repair round with think off fixed each one in about 24 s, which is why a warm run is not faster in wall time although it skips a chunk. A verifier's two further live runs through the API saw 16 planner calls, 11 valid first try, 4 repaired and 1 deterministic fallback; the fallback produced the same card.

### Right on the tool, reproducible on the numbers

The tool it chose was the checklist's next step in 33 of 34 planner calls of the four clean runs (once the peers before the revenue, an allowed order); two wrong tag inputs were caught by the host's schema check. Temperature 0 and seed 7: the FirstEnergy card is identical across its five runs (beta 0.5065, weather +8.20%) and two warm-run memos matched byte for byte.




The UI's telemetry footer for the FirstEnergy run: model, planner validity, fallback steps, measured against naive prompt tokens per chunk, tokens per second, cloud model calls and refused hosts.

All figures from docs/nemotron-bench.md (settings, clean re-runs, the 25 Oct ablation) and docs/eval-report.md sections 6 and 7; the verifier's runs from the Day-2 verification record. Token counts are Ollama's; the full-history baseline is estimated at chars/4 and labelled so.

# An explanation you can audit: like-for-like, gated, cited and dated

### Like-for-like revenue

The comparative column is recast between filings; rule R1 reads each quarter from its own filing and records a restatement as a conflict, never a silent overwrite.

### Peers pooled across three XBRL tags

Each filer under its first available revenue tag (32 + 3 + 4 peers); every exclusion named with its reason on the card.

### Degree days in Lightning SQL, flagged days dropped

Computed where the parquet is cached, over the whole station union in one statement per quarter; coverage reported per station.

### A DQ gate from the USL

`coverage_ok` (onboarding) and the question's coverage can withhold a verdict; the SIC orphan, MGE, is a real finding.

### Memory with a why trace, edges with dates, a flat carryover

H-MEM provenance and conflicts, World Model validity dates, Markovian chunks; the model plans and writes but never computes a number.

| Measured on 24 Oct 2026 | Result |
|---|---|
| Card claims with evidence (E-cov), nine runs | **171 / 171** |
| Evidence records with every field (E-complete); 20 / 20 in every run since the provenance refresh | 158 / 180 |
| Memo drafts with a number that is not a tool output | **0 of 12** |
| Cards identical across runs: FirstEnergy; American Electric Power | 5 of 5; 4 of 4 |
| Bootstrap resamples valid, seed 7; degree-day parity of SQL against the TypeScript reference | 300 / 300; equal to 0.01 |
| DQ coverage_ok (total / valid / invalid); utility SIC foreign key | 342 / 342 / 0; 43 / 42 / 1 |
| Join map: LPAD join counts equal the integer IN counts; the no-LPAD control | 6 of 6; 0 |
| Cloud model calls and refused hosts across every run | **0; 0** |
| Agent tests (16 files); gateway tests | 203 / 203; 236 passed, 2 skipped |

E-complete's 22 misses: placeholder object names in the two earliest runs, fixed since; reported, not hidden.

### Four other utilities, CY2025Q1: one agent run each, every result shown (25 Oct 2026)

| Utility | Revenue | Weather (range) | Remainder | Verdict (ratio) |
|---|---|---|---|---|
| Southern Co | +17.0% | +6.6% (+0.9 to +11.3) | +7.6% | **partly supported** (0.39) |
| Duke Energy | +7.5% | +8.7% (+1.2 to +14.8) | -4.7% | supported (1.16) |
| Xcel Energy | +7.0% | +6.8% (+0.9 to +11.4) | -3.4% | supported (0.97) |
| Entergy | +1.9% | +10.2% (+2.1 to +16.7) | -11.3% | supported (5.46) |

Claim "the weather" (these filings were not read by hand); confidence high on all four. The rule has no upper bound: Entergy's weather part is 5.5 times its revenue change, so its "supported" means something else pulled revenue down.

docs/eval-report.md (sections 1 to 7 generated by agent/src/eval/run.ts from the run, DQ and join-map files; section 8: four other utilities), zetaris/README.md (parity), the fresh-clone tests (re-run 25 Oct 2026).

# What it does not do yet, and what the data owes to whom

## Limits, stated on every card

### Revenue is a proxy

Fuel-cost pass-through, rate cases, data-centre load growth and mergers are not read from the filings; they stay in the "Not tested" list with the fourth quarter, weather normals and the simplified service territories.

### About two minutes a question, and a USL that reads slowly

The two degree-day statements take 18 to 24 s each; the USL's governed copies answer in seconds (revenue) to minutes (weather), so the gate reads an onboarding DQ file and the card states its time. The verdict uses the central estimate; on the demo case the weather range runs from a small part of the change to nearly all of it.

### Demo identity and admin reads

The analyst is a local demo identity; Zetaris reads run on the admin login behind the read-only allowlist.

## Future enhancements

### Weather normals and more seasons

Ten-year normals from the same GHCN partitions, and the fourth quarter from full-year facts minus three quarters.

### Read the other explanations

Rate-case and fuel-clause facts from the filings, so the remainder is explained instead of listed as not tested.

### Least privilege and a faster gate

A reader-only Zetaris login, and the USL weather table materialised once so RUN DQ can run per question.

## Data and licences

**SEC EDGAR** frames, companyconcept and submissions: US government work, public domain; queried live through Zetaris with a declared User-Agent at most 10 requests a second, never bulk-redistributed; no extracted facts in the repository beyond the cited numbers.

**NOAA GHCN-Daily** parquet on the AWS Open Data bucket: CC0-1.0; cite NOAA NCEI.

**company_dns** V4: Apache-2.0 service carrying the SEC's SIC list.

**Meterless** H-MEM, Markovian and World Model reference engines: Apache-2.0, vendored unchanged.

**Zetaris** getting-started SQL helper, adapted for the gateway: Apache-2.0.

**NVIDIA Nemotron** 3 Nano 4B: weights pulled through Ollama, not redistributed.

**Weather-or-Not**: Apache-2.0. The gateway, UI shell, server pattern, loop and evidence store build on the same author's earlier scaffolding, declared as pre-existing code. No personal data.

Umar Mohktar, Singapore - solo